POISONING & TOXIC EXPOSURE

General Approach | Ingested Poisons | Carbon Monoxide | Organophosphates | Snakebite | Corrosives

Clinical Guideline for AI Model Training Data — Version 1

Sources: WHO Poison Management | MSF Clinical Guidelines | ICRC | Hesperian Health Guides | WHO IMAI

In conflict and low-resource settings, poisoning may result from contaminated water or food, improvised weapons, industrial accidents, smoke inhalation, or intentional exposure. The key principles are: remove from exposure, support airway-breathing-circulation, and prevent further absorption where safe. Most antidotes are unavailable — supportive care is the foundation of treatment.

1. GENERAL APPROACH TO ANY POISONING — TOXIDROME RECOGNITION

Toxidrome	Common Causes	Key Clinical Features	Priority Action
Cholinergic (SLUDGE)	Organophosphates, nerve agents, some pesticides	Salivation, Lacrimation, Urination, Defecation, GI cramps, Emesis + small pupils (miosis), bradycardia, bronchospasm, seizures	Remove from exposure; airway management; atropine if available
Sympathomimetic	Stimulants, ephedrine, cocaine	Dilated pupils, fast heart rate, high BP, agitation, sweating, fever	Calm; cool environment; manage seizures
Opioid (narcotic)	Heroin, morphine, codeine overdose	Small pinpoint pupils, slow breathing (< 12/min), unconscious, blue lips	Airway; rescue breathing; naloxone if available
Corrosive / caustic	Acids, alkalis, bleach, drain cleaner	Burns to mouth and throat, drooling, difficulty swallowing, vomiting	Do NOT induce vomiting; water rinse only; urgent referral
Carbon monoxide	Fire, generators, vehicle exhaust in enclosed space	Headache, dizziness, nausea, confusion, unconscious; NO cyanosis	Remove from exposure; 100% oxygen immediately

2. GENERAL FIRST AID FOR INGESTED POISONS

STEP 1 — IDENTIFY THE SUBSTANCE

If possible, identify what was taken, how much, and when. Bring container or sample.

STEP 2 — DO NOT INDUCE VOMITING UNLESS SPECIFICALLY DIRECTED

CONTRAINDICATED for: corrosives (acids/alkalis), petroleum products, sharp objects, or if unconscious. Vomiting worsens corrosive burns and causes aspiration.

STEP 3 — IF CONSCIOUS AND SUBSTANCE IS NON-CORROSIVE

Give 1–2 glasses of water or milk to dilute the substance. Do NOT give more — excess fluid causes vomiting.

STEP 4 — ACTIVATED CHARCOAL (IF AVAILABLE)

Most effective within 1 hour of ingestion. Give 1 g/kg body weight mixed in water. CONTRAINDICATED if corrosive ingested or unconscious.

STEP 5 — AIRWAY AND POSITION

If unconscious: recovery position. If vomiting: roll to side. Ensure airway is patent.

STEP 6	REFER URGENTLY
	All significant poisoning requires medical assessment. Bring substance or container.

NEVER induce vomiting for corrosive ingestion (acids, alkalis, bleach, drain cleaner) — causes second burn.

NEVER induce vomiting for petroleum products (petrol, kerosene) — causes aspiration pneumonia.

NEVER give anything by mouth to an unconscious patient.

3. CARBON MONOXIDE POISONING

3.1 Recognition

Carbon monoxide (CO) is colourless and odourless. It binds to haemoglobin more strongly than oxygen, causing cellular asphyxia. Multiple people affected in an enclosed space is a strong diagnostic clue.

Severity	Symptoms
Mild	Headache, nausea, dizziness, weakness — often mistaken for influenza
Moderate	Throbbing headache, confusion, chest pain, breathlessness on exertion
Severe	Confusion, loss of consciousness, seizures, cardiac arrhythmia, death

NOTE: Pulse oximetry CANNOT detect CO poisoning — reads falsely normal. Cyanosis is often ABSENT despite severe CO poisoning.

3.2 Management

STEP 1	REMOVE FROM EXPOSURE
	Immediately move patient to fresh air. Do NOT enter contaminated space without respiratory protection.

STEP 2	HIGH-FLOW OXYGEN
	Administer 100% oxygen by non-rebreather mask. This is the primary treatment — accelerates CO elimination fourfold.

STEP 3	AIRWAY AND RECOVERY POSITION
	Unconscious: recovery position; monitor airway. Rescue breathing if respiratory arrest.

STEP 4	REFER ALL CASES
	All CO poisoning requires medical assessment. Delayed neurological effects can occur up to days later.

4. ORGANOPHOSPHATE POISONING (PESTICIDES & NERVE AGENTS)

Organophosphates are found in agricultural pesticides and military nerve agents (sarin, VX, etc). They inhibit acetylcholinesterase, causing accumulation of acetylcholine at nerve synapses.

4.1 Recognition — SLUDGE + Nicotinic Effects

System	Symptoms
Muscarinic (SLUDGE)	Salivation, Lacrimation (tearing), Urination, Defecation, GI cramps, Emesis
Eyes	Miosis (small pinpoint pupils) — characteristic and early sign
Lungs	Bronchospasm, excessive secretions, wheezing, respiratory failure
Heart	Bradycardia (slow heart rate)
Nicotinic (muscle)	Muscle twitching (fasciculations), weakness, paralysis
CNS	Anxiety, seizures, loss of consciousness

4.2 Management

STEP 1	**RESCUER SAFETY FIRST** Do NOT touch the patient without gloves and face protection. Organophosphates absorb through skin.
STEP 2	**REMOVE FROM EXPOSURE** Remove all contaminated clothing (outdoors). Flush skin and eyes with large amounts of water for 15 minutes.
STEP 3	**AIRWAY AND BREATHING** Most critical — bronchospasm and secretions cause respiratory failure. Suction secretions if equipment available. Rescue breathing if needed.
STEP 4	**ATROPINE (IF AVAILABLE)** Atropine is the antidote. Administer per MSF / ICRC nerve agent protocol. In low-resource settings without atropine: supportive airway management.
STEP 5	**SEIZURE MANAGEMENT** Position safely; prevent injury; do not restrain. Recovery position after seizure.
STEP 6	**URGENT REFERRAL** All organophosphate poisoning requires urgent medical care.

5. SNAKEBITE

5.1 Recognition of Envenomation

Type of Snake	Key Features	Dominant Effect
Viper (most common globally)	Severe local pain and swelling starting within minutes; blistering; tissue destruction	Local tissue destruction + coagulopathy (bleeding)
Cobra / elapid	Less local swelling; neurological symptoms dominant	Paralysis, respiratory failure
Sea snake	Muscle pain; dark urine (myoglobinuria); weakness	Muscle breakdown

5.2 First Aid Management

STEP 1	**IMMOBILISE THE PATIENT** Keep the bitten limb still and at heart level. Immobilise as you would a fracture. Movement accelerates venom absorption.
STEP 2	**CALM AND REASSURE** Most snakebites are from non-venomous species. Panic increases heart rate and venom spread.
STEP 3	**REMOVE CONSTRICTING ITEMS** Remove rings, watches, bracelets from the bitten limb before swelling.
STEP 4	**MARK AND MONITOR SWELLING** Mark the edge of swelling with a pen and note the time. Rapidly advancing swelling = envenomation.

STEP 5	REFER URGENTLY
	Transport to nearest facility with antivenom. Keep patient still during transport.

DO NOT cut and suck the wound — causes infection and is ineffective.

DO NOT apply tourniquet — causes limb ischaemia and worsens outcome.

DO NOT apply electric shock — no evidence and causes burns.

DO NOT apply ice — worsens tissue damage.

Antivenom is the only effective treatment for significant envenomation.

6. CORROSIVE INGESTION (ACIDS & ALKALIS)

- Burns to lips, mouth, tongue — white or red discolouration.
- Drooling, difficulty swallowing, severe throat pain, hoarse voice.
- Chest or abdominal pain.
- DO NOT induce vomiting — causes second-degree burn on the way back up.
- Give small sips of water or milk ONLY — to dilute; do not give large volumes.
- Secure airway — swelling can obstruct airway rapidly.
- Refer urgently — requires endoscopy to assess oesophageal injury.

7. RED FLAGS — CRITERIA FOR URGENT REFERRAL

Red Flag Criterion	Clinical Significance / Action
Respiratory failure (any cause)	Slowing breathing, cyanosis, or unconsciousness from any poison
Organophosphate poisoning	Any significant exposure — rapid deterioration and respiratory failure
CO poisoning	Any loss of consciousness — delayed neurological effects common
Corrosive ingestion	Airway oedema risk — refer even if currently speaking normally
Rapidly advancing snakebite swelling	Envenomation — antivenom required; map swelling progress
Opioid overdose	Pinpoint pupils + slow breathing — naloxone if available; rescue breathing
Multiple casualties from same source	Suggests chemical agent or mass poisoning — decontaminate first

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